

Phase 3 Documentation

Project Title: SmartCare CRM – Healthcare Appointment & Patient Tracking System

Phase 3: Data Modeling & Relationships

1. Introduction

In this phase, we focused on building the data model for SmartCare CRM. This included creating standard and custom objects, defining fields, setting up relationships (Lookup, Master-Detail, Hierarchical, and Junction), and designing record types and layouts. Schema Builder was used to visually validate and refine relationships, ensuring the data structure supports key healthcare processes like patient management, appointments, and prescriptions.

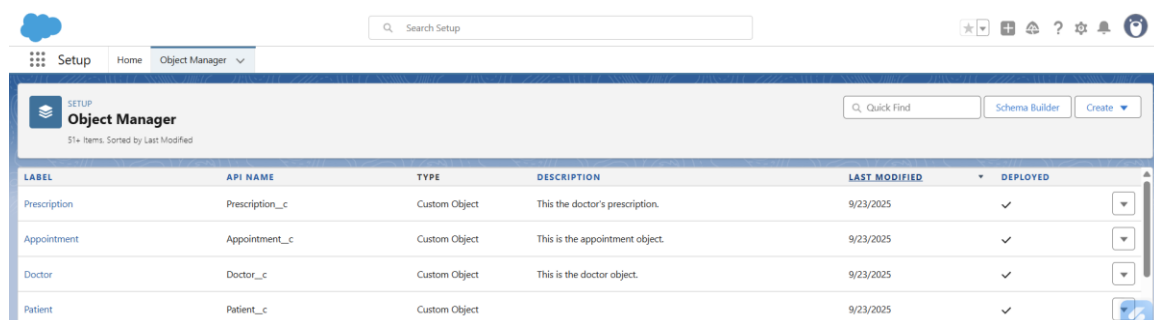
2. Objectives

- To design a scalable and normalized data model.
- To implement relationships between core entities (Patients, Doctors, Appointments, Prescriptions).
- To configure record types, page layouts, and compact layouts for role-specific usability.
- To explore and demonstrate use of junction objects and external objects.
- To validate the model using Schema Builder.

3. Implementation Steps

A. Standard & Custom Objects

- Standard Objects Used: User, Contact.
- Custom Objects Created: Patient_c, Appointment_c, Prescription_c, Doctor_c.



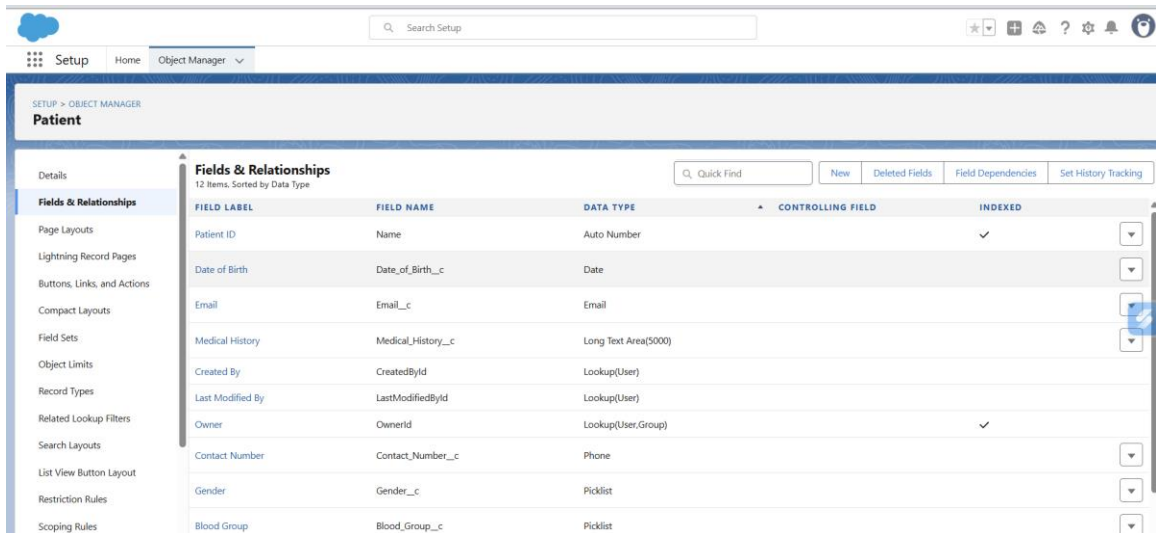
The screenshot displays the Salesforce Object Manager interface. At the top, there is a navigation bar with tabs for Setup, Home, and Object Manager. Below the navigation bar, the Object Manager header includes a search bar, a 'Quick Find' button, and a 'Schema Builder' button. The main content area shows a table of custom objects. The table has columns for Label, API Name, Type, Description, Last Modified, and Deployed. The objects listed are Prescription, Appointment, Doctor, and Patient, all of which are Custom Objects and have been deployed.

LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Prescription	Prescription_c	Custom Object	This the doctor's prescription.	9/23/2025	✓
Appointment	Appointment_c	Custom Object	This is the appointment object.	9/23/2025	✓
Doctor	Doctor_c	Custom Object	This is the doctor object.	9/23/2025	✓
Patient	Patient_c	Custom Object		9/23/2025	✓

B. Fields

Each object was enriched with fields:

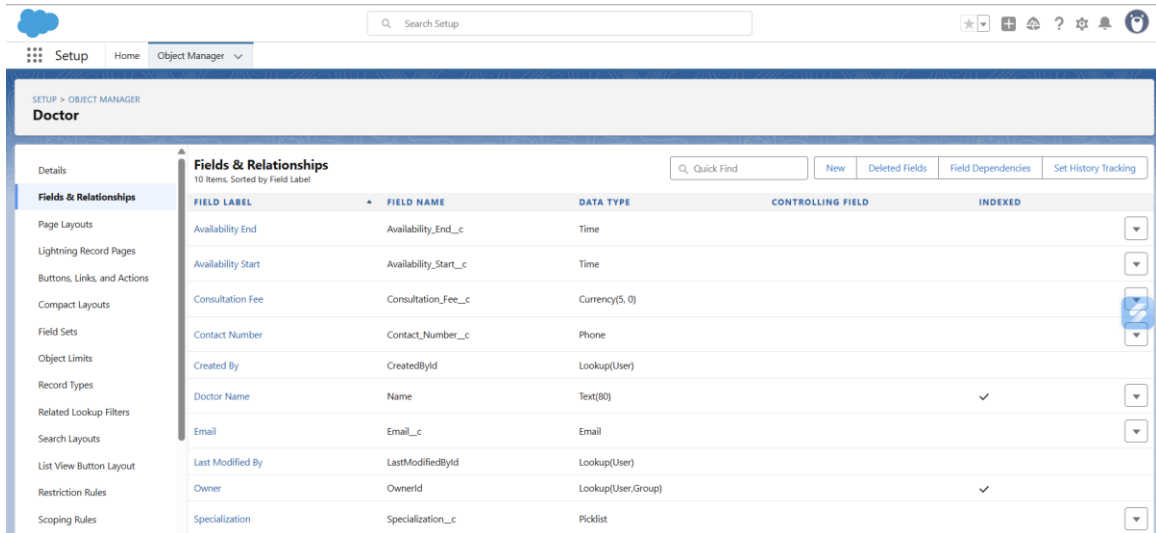
- Patient_c:



The screenshot shows the 'Patient' object configuration in Salesforce. The 'Fields & Relationships' section lists 12 fields. The table below represents the data shown in the interface.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Patient ID	Name	Auto Number		✓
Date of Birth	Date_of_Birth__c	Date		
Email	Email__c	Email		
Medical History	Medical_History__c	Long Text Area(5000)		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Contact Number	Contact_Number__c	Phone		
Gender	Gender__c	Picklist		
Blood Group	Blood_Group__c	Picklist		

- Doctor_c: Specialization, Contact, Availability.



The screenshot shows the 'Doctor' object configuration in Salesforce. The 'Fields & Relationships' section lists 10 fields. The table below represents the data shown in the interface.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Availability End	Availability_End__c	Time		
Availability Start	Availability_Start__c	Time		
Consultation Fee	Consultation_Fee__c	Currency(5, 0)		
Contact Number	Contact_Number__c	Phone		
Created By	CreatedById	Lookup(User)		
Doctor Name	Name	Text(80)		✓
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Specialization	Specialization__c	Picklist		

- Appointment_c: Appointment Date/Time, Status, Visit Type, Assigned Doctor.

Setup > Object Manager

Appointment

Fields & Relationships

12 Items. Sorted by Field Label

Field Label	Field Name	Data Type	Controlling Field	Indexed
Appointment DateTime	Appointment_DateTime_c	Date/Time		
Appointment Number	Name	Auto Number		✓
Assigned Doctor	Assigned_Doctor_c	Lookup(Doctor)		✓
Consultation Fee	Consultation_Fee_c	Currency(4, 0)		
Created By	CreatedById	Lookup(User)		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User, Group)		✓
Patient	Patient_c	Lookup(Patient)		✓
Reason	Reason_c	Long Text Area(32768)		
Record Type	RecordTypeId	Record Type		✓
Status	Status_c	Picklist		

- Prescription_c: Medications, Dosage, Linked Appointment.

Setup > Object Manager

Prescription

Fields & Relationships

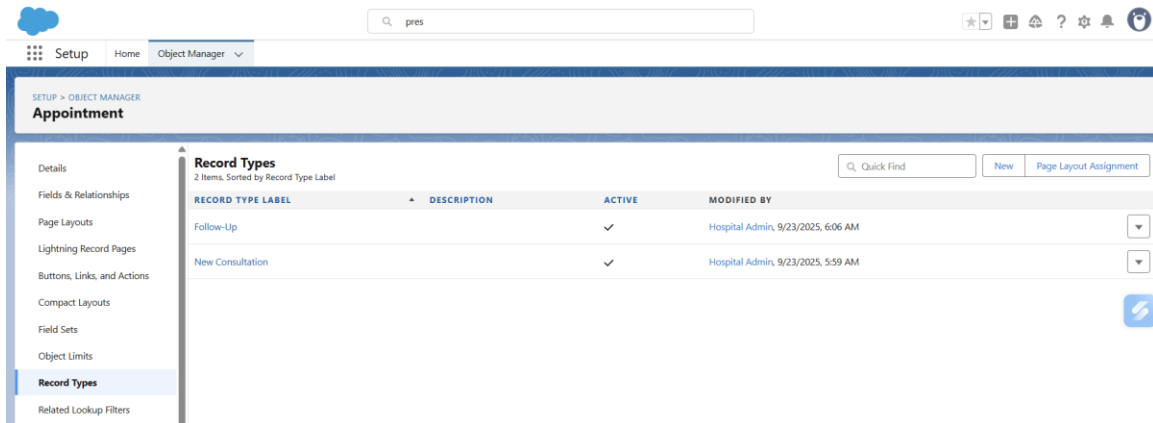
8 Items. Sorted by Field Label

Field Label	Field Name	Data Type	Controlling Field	Indexed
Appointment	Appointment_c	Master-Detail(Appointment)		✓
Created By	CreatedById	Lookup(User)		
Dosage Instructions	Dosage_Instructions_c	Long Text Area(32768)		
Follow Up Date	Follow_Up_Date_c	Date		
Last Modified By	LastModifiedById	Lookup(User)		
Medications	Medications_c	Long Text Area(300)		
Notes	Notes_c	Long Text Area(32768)		
Prescription Number	Name	Auto Number		✓

C. Record Types

Appointment_c Record Types:

- New Consultation
- Follow-Up

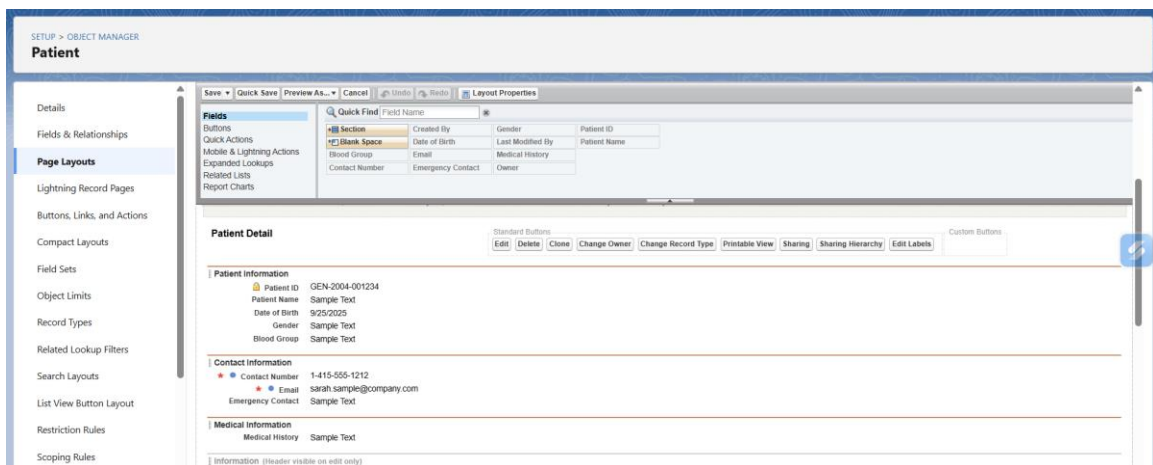


The screenshot shows the Salesforce Setup interface for the 'Appointment' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, and Related Lookup Filters. The main content area is titled 'Record Types' and shows a table with 2 items, sorted by Record Type Label. The table has columns for Record Type Label, Description, Active status, and Modified By.

RECORD TYPE LABEL	DESCRIPTION	ACTIVE	MODIFIED BY
Follow-Up		✓	Hospital Admin, 9/23/2025, 6:06 AM
New Consultation		✓	Hospital Admin, 9/23/2025, 5:59 AM

D. Page Layouts

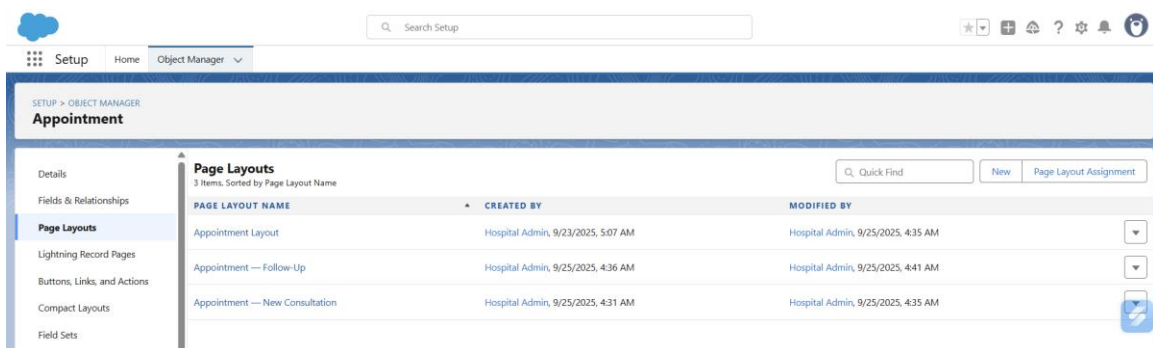
Patient Layout: Patient Info, Contact Info, Medical History, Related Lists.



The screenshot shows the Salesforce Setup interface for the 'Patient' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Patient Detail' and shows a preview of the patient layout. The layout includes sections for Patient Information, Contact Information, and Medical Information, each with a list of fields and their values.

Section	Field Name	Value
Patient Information	Patient ID	GEN-2004-001234
	Patient Name	Sample Text
	Date of Birth	9/25/2025
	Gender	Sample Text
	Blood Group	Sample Text
Contact Information	Contact Number	1-415-555-1212
	Email	sarah.sample@company.com
	Emergency Contact	Sample Text
Medical Information	Medical History	Sample Text

Appointment Layouts: New Consultation (fee, reason, doctor), Follow-Up (notes).



The screenshot shows the Salesforce Setup interface for the 'Appointment' object. The left sidebar contains a navigation menu with options like Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, Record Types, Related Lookup Filters, Search Layouts, List View Button Layout, Restriction Rules, and Scoping Rules. The main content area is titled 'Page Layouts' and shows a table with 3 items, sorted by Page Layout Name. The table has columns for Page Layout Name, Created By, and Modified By.

PAGE LAYOUT NAME	CREATED BY	MODIFIED BY
Appointment Layout	Hospital Admin, 9/23/2025, 5:07 AM	Hospital Admin, 9/25/2025, 4:35 AM
Appointment — Follow-Up	Hospital Admin, 9/25/2025, 4:36 AM	Hospital Admin, 9/25/2025, 4:41 AM
Appointment — New Consultation	Hospital Admin, 9/25/2025, 4:31 AM	Hospital Admin, 9/25/2025, 4:35 AM

Prescription Layout: Prescription details + related Appointment.

E. Compact Layouts

- Patient Compact Layout: Patient Name, Age, Gender, Assigned Doctor.

